THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 12 Solution
2026/04/22 and 2026/04/24

1. Let $\{X_n\}_{n\geq 0}$ be a Markov chain with state space S. Show that for any $n \geq 1$ and any $x_0, x_1, \cdots, x_n \in S$,

$$P(X_0 = x_0 \mid X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n) = P(X_0 = x_0 \mid X_1 = x_1).$$

Solution:

By the Markov property, we have

$$\begin{aligned}
&P\left(X_0 = x_0 \mid X_1 = x_1, \cdots, X_n = x_n\right) \\
&= \frac{P\left(X_0 = x_0, X_1 = x_1, \cdots, X_n = x_n\right)}{P\left(X_1 = x_1, \cdots, X_n = x_n\right)} \\
&= \frac{P\left(X_0 = x_0\right) P\left(X_1 = x_1 \mid X_0 = x_0\right) \cdots P\left(X_n = x_n \mid X_{n-1} = x_{n-1}, \cdots, X_0 = x_0\right)}{P\left(X_1 = x_1\right) P\left(X_2 = x_2 \mid X_1 = x_1\right) \cdots P\left(X_n = x_n \mid X_{n-1} = x_{n-1}, \cdots, X_1 = x_1\right)} \\
&= \frac{P\left(X_0 = x_0\right) p_{0,1} \cdots p_{n-1,n}}{P\left(X_1 = x_1\right) p_{1,2} \cdots p_{n-1,n}} \\
&= \frac{P\left(X_0 = x_0\right) p_{0,1}}{P\left(X_1 = x_1\right)} \\
&= \frac{P\left(X_0 = x_0, X_1 = x_1\right)}{P\left(X_1 = x_1\right)} \\
&= P\left(X_0 = x_0 \mid X_1 = x_1\right)
\end{aligned}$$

2. Let $\{X_n\}_{n\geq 0}$ be a branching process defined by $X_{n+1} = \sum_{i=1}^{X_n} \xi_i^{(n+1)}$, where $\{\xi_i^{(n)}\}$ are i.i.d. random variables independent of X_0. Suppose that each $\xi_i^{(n)}$ has mean μ and variance σ^2. Show that

$$\mathbb{E}[X_{n+1}^2 \mid X_n = m] = m\sigma^2 + m^2\mu^2.$$

Solution:

By the i.i.d. property of $\left\{\xi_i^{(n)}\right\}$, we have

$$\begin{aligned}
\mathbb{E}\left[X_{n+1}^2 \mid X_n = m\right] &= \mathbb{E}\left[\left(\sum_{i=1}^{m} \xi_i^{(n+1)}\right)^2\right] \\
&= \operatorname{Var}\left(\sum_{i=1}^{m} \xi_i^{(n+1)}\right) + \left(\mathbb{E}\left[\sum_{i=1}^{m} \xi_i^{(n+1)}\right]\right)^2 \\
&= m\sigma^2 + m^2\mu^2.
\end{aligned}$$

3. Let $\{X_n\}_{n\geq 0}$ be the Ehrenfest chain with state space $\{0, 1, \cdots, d\}$. Suppose that X_0 has a binomial distribution with parameters d and $1/2$, i.e.,

$$P(X_0 = x) = \frac{1}{2^d} \cdot \binom{d}{x}, \quad x = 0, \cdots, d.$$

Find the distribution of X_1.

Solution:

Recall that the transition probability is given by

$$p_{x,y} = \begin{cases} \frac{x}{d}, & y = x - 1, \\ \frac{d-x}{d}, & y = x + 1. \end{cases}$$

Hence,

$$P(X_1 = 0) = P(X_0 = 1)\, p_{1,0} = \frac{d}{2^d} \cdot \frac{1}{d} = \frac{1}{2^d},$$
$$P(X_1 = d) = P(X_0 = d - 1)\, p_{d-1,d} = \frac{d}{2^d} \cdot \frac{1}{d} = \frac{1}{2^d},$$

and for $x \in \{1, \cdots, d-1\}$,

$$\begin{aligned} P(X_1 = x) &= P(X_0 = x - 1)\, p_{x-1,x} + P(X_0 = x + 1)\, p_{x+1,x} \\ &= \frac{\binom{d}{x-1}}{2^d} \cdot \frac{d - x + 1}{d} + \frac{\binom{d}{x+1}}{2^d} \cdot \frac{x + 1}{d} \\ &= \frac{d!}{(d-x)!(x-1)!} \cdot \frac{1}{d \cdot 2^d} + \frac{d!}{(d-x-1)!x!} \cdot \frac{1}{d \cdot 2^d} \\ &= \frac{x\binom{d}{x} + (d-x)\binom{d}{x}}{d \cdot 2^d} \\ &= \frac{\binom{d}{x}}{2^d}. \end{aligned}$$

Therefore, the distribution of X_1 is still the binomial distribution with parameters d and $1/2$.

4. Let $\{X_n\}_{n\geq 0}$ be a Markov chain whose state space S is a subset of $\{0, 1, 2, \cdots\}$. Suppose the transition probability satisfies

$$\sum_{y \in S} y p_{x,y} = Ax + B, \quad x \in S$$

for some constants A and B.

(a) Show that $\mathbb{E}[X_{n+1}] = A\mathbb{E}[X_n] + B$.

(b) Show that if $A \neq 1$, then

$$\mathbb{E}[X_n] = \frac{B}{1 - A} + A^n \left(\mathbb{E}[X_0] - \frac{B}{1 - A} \right)$$

(c) Let $\{X_n\}_{n\geq 0}$ be the Ehrenfest chain on $\{0, 1, \cdots, d\}$ with $X_0 = x$. Compute $\mathbb{E}[X_n]$.

Solution:

(a) Using the formula of total probability

$$\begin{aligned} E[X_{n+1}] = \mathbb{E}[\mathbb{E}[X_{n+1} \mid X_n]] &= \sum_{x\in S} P(X_n = x) \sum_{y\in S} yP\left(X_{n+1} = y \mid X_n = x\right) \\ &= \sum_{x\in S} P(X_n = x) \sum_{y\in S} yp_{x,y} \\ &= \sum_{x\in S} P(X_n = x)(Ax + B) \\ &= AE[X_n] + B \end{aligned}$$

(b) If $A \neq 1$, then from (a) we know

$$\mathbb{E}[X_{n+1}] - \frac{B}{1-A} = A\mathbb{E}[X_n] + B - \frac{B}{1-A} = A\left(\mathbb{E}[X_n] - \frac{B}{1-A}\right)$$

Inductively we have

$$\mathbb{E}[X_n] - \frac{B}{1-A} = A^n\left(\mathbb{E}[X_0] - \frac{B}{1-A}\right)$$

which leads to the result.

(c) For the Ehrenfest chain on $S = \{0, 1, \cdots, d\}$, one can check that

$$\sum_{y\in S} yp_{x,y} = (x-1)\frac{x}{d} + (x+1)\frac{d-x}{d} = \frac{d-2}{d}x + 1, \quad \forall x \in S.$$

Note in particular that the above also holds for $x = 0$ and $x = d$. Applying the result in part (b) with $A = \frac{d-2}{d}$, $B = 1$, and $\mathbb{E}[X_0] = x$, we have

$$\mathbb{E}\left[X_n\right] = \frac{d}{2} + \left(\frac{d-2}{d}\right)^n \left(x - \frac{d}{2}\right).$$